

Card Transaction Fraud Data Preparation Report

Part 1: Data Cleaning Process

1. Outlier Handling

- **Extreme Amount Filtering:** Removed 1 transaction with Amount > 3,000,000 (non-fraudulent Mexican retail purchase in pesos).
- **FedEx Microtransactions:** Flagged 7,592 transactions from FEDEX with amounts between \$3.62–\$3.80 using amount_okay but retained them in the dataset.

2. Exclusions

- **Transaction Type:** Excluded 355 non-purchase transactions (types A, D, Y), retaining only type P (97,497 transactions).

3. Imputation

Merchnum (Merchant Number)

- Replaced '0' with NaN and imputed 1,164 missing values via Merch description mappings.
- Generated new unique IDs for 515 unmatched descriptions (e.g., "RETAIL CREDIT ADJUSTMENT" → "unknown").

Merch State

- Filled missing states hierarchically:
 - a. Merch zip mappings (primary)
 - b. Merchnum/Merch description mappings (secondary)
 - c. Categorized non-U.S. states as "foreign" and unresolved cases as "unknown".

Merch Zip

- Imputed missing zips using:
 - a. Merchnum/Merch description mappings
 - b. Most populous zip code for the state if unresolved

- c. Defaulted to "unknown" for adjustments.

Part 2: Variable Creation Summary

Summary Table of Created Variables

Variable Group	# Variables	Description
Target Encoding (TE)	3	Smoothed fraud risk scores for Merch state, Merch zip, and Dow using historical fraud rates.
Entity Interaction Variables	148	Composite keys (e.g., card_merch, card_zip) to track relationships between entities (card, merchant, zip, etc.).
Days Since Variables	23	Days since last transaction for each entity (e.g., Cardnum_day_since, Merchnum_day_since).
Transaction Count Variables	1,449	Number of transactions per entity over 0/1/3/7/14/30/60-day windows.
Amount Statistics Variables	644	Mean/max/median/total transaction amounts per entity over rolling windows.
Amount Ratio Variables	644	Ratios of current amount to historical metrics (e.g., actual/avg, actual/max).
Velocity Change Variables	368	Short-term vs. long-term activity ratios (e.g., count_1 / count_7).
Variability Variables	414	Mean/max/median deviation in transaction amounts over time windows.
Unique Entity Count Variables	230	Unique secondary entities (e.g., zips) linked to primary entities (e.g., cards).
Geospatial Variables	7	Distance between consecutive transactions and flags for implausible distances (>1,000 miles).
Temporal Variables	5	Weekend indicator (Is_Weekend), hour of day, and time since midnight.
Behavioral Flags	12	High amount for merchant, state inconsistency, merchant dominance score.
Foreign Transaction Variables	3	Indicators for non-U.S. merchant zips.
Benford's Law Features	2	Anomaly detection in transaction amount digit distributions (excluded due to overfitting).

Miscellaneous Features	428	Derived metrics (e.g., velocity ratios normalized by days since, interaction variables).
------------------------	-----	--

Key Variable Group Descriptions

1. Target Encoding (TE):

- Converts categorical fields into fraud risk scores using logistic smoothing on historical data.
- Trained only on pre-2010-11-01 data to prevent temporal leakage.

2. Entity Interaction Variables:

- Created by concatenating entity IDs (e.g., `card_merch = Cardnum + Merchnum`) to track behavioral patterns across linked entities.

3. Time-Based Aggregates:

- Rolling window statistics (0/1/3/7/14/30/60 days) for counts, amounts, and ratios.
- Examples: `card_merch_count_7`, `Merchnum_avg_30`.

4. Velocity Change Variables:

- Detect sudden spikes in activity (e.g., `count_1 / count_7 > 3` indicates rapid activity increase).

5. Geospatial Variables:

- Computed haversine distance between consecutive merchant zips using `geopy`.
- Flags transactions with implausible geographic jumps (>1,000 miles within 24 hours).

6. Behavioral Flags:

- `Is_Weekend`: Weekend transaction indicator.
- `State_inconsistency`: Card used in multiple states within 24 hours.
- `Merchant_dominance_score`: Fraction of a card's transactions at its most frequent merchant.

Design Principles

- **Temporal Integrity:** Variables for a transaction use only prior data (no future-looking aggregates).

- **Hierarchical Imputation:** Leveraged relationships between fields (e.g., zip → state) to maximize data utility.
- **Smoothing:** Applied to target encoding and Benford's Law metrics to stabilize rare categories.

This pipeline transformed raw transaction data into a structured dataset optimized for real-time fraud detection.

New Variables for Fraud Detection

Variable Name	Description	Logic
distance_from_last_transaction	Haversine distance (miles) between current and previous transaction for the same card.	Calculated using geopy with consecutive merchant zips. Missing zips set to 0.
Is_Weekend	Binary flag (1/0) for transactions occurring on Saturday/Sunday.	Derived from Date using <code>datetime.weekday()</code> .
Hour_of_Day	Hour of transaction (0-23) as a numeric feature.	Extracted from Date using <code>datetime.hour</code> .
velocity_1h_vs_24h	Ratio of transactions in the last 1 hour to the last 24 hours for the same card.	$(\text{count_1h} + 1) / (\text{count_24h} + 1)$ to avoid division by zero.
amount_deviation_from_avg	Absolute difference between current amount and cardholder's 30-day average.	<code>abs(Amount - Cardnum_avg_30)</code> .
new_merchant_flag	Binary flag (1/0) indicating if the merchant has never been transacted by the cardholder before.	1 if <code>Merchnum</code> not in card's historical <code>Merchnum</code> list prior to current transaction, else 0.
cross_border_24h	Binary flag (1/0) if the card was used in a different country within 24 hours. Non-U.S. <code>Merch state</code> categorized as 'foreign'.	Check if <code>Merch state</code> differs from prior transaction's state and one is 'foreign' within 24 hours.
amount_rounded	Binary flag (1/0) for transactions where the amount is a whole number (e.g., \$20.00).	1 if <code>Amount % 1 == 0</code> , else 0.
minutes_since_last_tx	Minutes elapsed since the cardholder's previous transaction.	$(\text{Current transaction time} - \text{Last transaction time}).\text{total_seconds}() / 60$.
high_risk_category_ratio	Percentage of a cardholder's transactions in high-risk merchant categories (e.g., jewelry, electronics) over the past 90 days.	$(\text{count_high_risk_90} / \text{count_total_90})$. High-risk categories predefined via external merchant code lists.